

SA-GS: Season-aware affine 3D Gaussian Splatting for satellite image rendering

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ARTICLE INFO

Keywords:

3D Gaussian Splatting
Affine camera
Satellite image rendering
View augmentation

ABSTRACT

Recently, 3D Gaussian Splatting (3DGS) has shown considerable potential in satellite image rendering. However, this technique was originally explored under the pinhole camera model, leading to its limited capability of handling satellite images that are generally captured by line-array sensors but not in accord with the pinhole camera model. To address this problem, we propose a Season-aware Affine 3D Gaussian Splatting method (called SA-GS) for rendering line-array satellite images under the affine camera model, motivated by the fact that the local regions of line-array images could be effectively modeled by the affine camera model. Specifically, the proposed method consists of an affine 3D Gaussian Splatting module, a seasonal embedding module, and an affine view augmentation module. In the affine 3D Gaussian Splatting module, an affine version of 3D Gaussian Splatting is explored according to the affine camera model. Then, the seasonal embedding module is designed to inject season embeddings into the Gaussian representations of the aforementioned module, so that the learnt 3D Gaussians could tolerate significant textural variations among cross-season satellite images. In addition, the affine view augmentation module is designed to generate synthetic satellite images, which could provide additional supervision for further improving the performance of SA-GS. Extensive experimental results on 8 areas of interest from the DFC2019 dataset demonstrate that the proposed SA-GS obtains an average SSIM of 0.702, PSNR of 22.86 dB, and LPIPS of 0.311, significantly outperforming 11 state-of-the-art methods for satellite image rendering, including Sat-NeRF, EOGS, SatGS, etc. The code is available at <https://github.com/CosyXu/SA-GS>.

1. Introduction

Novel view rendering is a challenging task in remote sensing (Wu et al., 2022; Lv et al., 2023; Ren et al., 2024) and computer vision (Xu and Qin, 2025; K. Chen et al., 2024; Wei et al., 2024). Most of the existing methods in literature (Marí et al., 2022; Zhang and Rupnik, 2023; Sun et al., 2024) are based on Neural Radiance Fields (NeRF) (Mildenhall et al., 2021). However, these methods generally require time-consuming training and rendering. Recently, 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) has shown its usefulness and potential in several satellite-imagery-related tasks as verified by some existing works in literature, such as digital elevation modeling (Aira et al., 2025), remote sensing multi-view segmentation (Qi et al., 2025), and satellite image rendering (Bai et al., 2025). Moreover, 3DGS has shown its superior capability of rendering optical plane-array images at a much faster speed. As indicated in T. Zhang et al. (2024) and X. Zhou et al. (2024), fast novel view synthesis plays an important role in several practical remote sensing applications, such as the large-scale

urban and landscape monitoring task and the change detection task, where fast rendering from arbitrary viewpoints facilitates interactive 3D scene monitoring and enables faster analysis and more responsive decision-making. These applications motivate us to investigate a 3DGS-based fast novel view synthesis method for satellite images that are generally captured by line-array sensors.

In fact, for most of the existing 3DGS-based rendering methods that are explored under the pinhole camera model (Y. Chen et al., 2024; Charatan et al., 2024; Lu et al., 2024), they are generally available for handling plane-array images, but perform poorly on line-array satellite images that are not subject to the pinhole camera model, also as demonstrated by the results in Tables 2 and 3 of this paper. It is further noted that a few works (Wang et al., 2024; Bai et al., 2025) employ additional constraints or operations for improving their capability of satellite image rendering; however, the improvements are still limited due to their dependence on the pinhole camera model.

To address the above problem, we propose a Season-aware Affine 3D Gaussian Splatting method for rendering line-array satellite images

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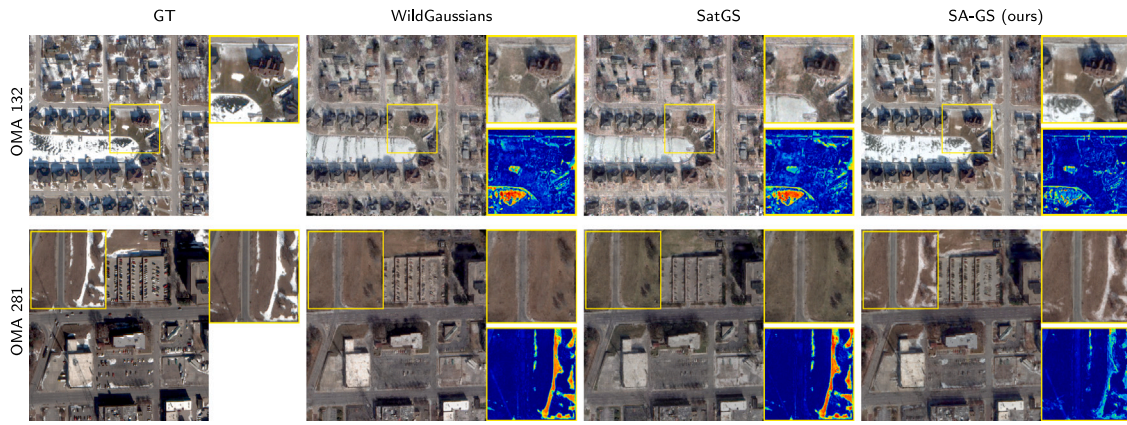


Fig. 1. Visualization results of WildGaussians, SatGS, and the proposed SA-GS on AOIs “OMA 132” and “OMA 163” from the DFC2019 dataset (Le Saux et al., 2019). For each visualization result of the comparative methods, the sub-figure at the top right corner is the enlarged region corresponding to the highlighted yellow rectangle of the left rendered image, while the sub-figure at the bottom right corner is the enlarged error map corresponding to the highlighted yellow rectangle of the left rendered image. In each error map, bluer indicates a smaller error, while redder indicates a larger error.

under the affine camera model, called SA-GS, motivated by the fact that local regions of line-array images could be effectively modeled by the affine camera model (Hartley and Zisserman, 2003). Specifically, the proposed SA-GS contains an affine 3D Gaussian Splatting module, a seasonal embedding module, and an affine view augmentation module. The affine 3D Gaussian Splatting module serves as the rendering pipeline of SA-GS, and an affine version of 3D Gaussian Splatting (Kerbl et al., 2023) is explored under the affine camera model in this module, where the affine projection is employed to replace the standard perspective projection. It has to be explained that in EOGS (Aira et al., 2025), the affine projection is also employed to adapt 3DGS to satellite imagery. The main difference between the two methods is that they use different manners to estimate the affine projection parameters. Specifically, in EOGS, the affine projection is obtained by approximating the RPC (rational polynomial camera) model that is estimated according to the sensor data. This means that EOGS has to suffer from an accumulated error problem in principle, caused by the accumulation of both the RPC error and the affine parameter estimation error. In contrast, the affine projection in our method is estimated directly from the training data, which not only requires the RPC model, but also avoids the accumulated error problem.

Then, it is noted that for a given scene, the images captured in different seasons generally show significantly different colors and textures. If these images are used together to train a 3DGS-based rendering method without additional operations, each Gaussian representation has to be required to represent various colors and textures, leading to ambiguous rendering results, which could be illustrated by Fig. 1. As seen from this figure, the local regions in the highlighted yellow rectangles of the ground-truth images in Column 1 are covered by snow, but the corresponding local regions synthesized by two existing SOTA 3DGS-based methods (WildGaussians (Kulhanek et al., 2024) and SatGS (Bai et al., 2025)) either show no snow coverage or blur snow regions as shown in the second and third columns. This issue motivates us to introduce the seasonal embedding module, where a set of season embedding vectors is injected into each Gaussian representation for discriminating color and texture differences caused by seasonal changes. The visualization results in the fourth column of Fig. 1 show that the proposed SA-GS could effectively render images corresponding to different seasons.

Also, as indicated in Derksen and Izzo (2021) and Zhang and Rupnik (2023), the number of satellite images in many remote sensing scenarios is generally limited (taking AOI “JAX 004” from the DFC2019 dataset (Le Saux et al., 2019) for example, there are only 7 training images). It is further noted from other works (Xiong et al., 2023; Han et al., 2024) that when a 3DGS-based method is trained with a small number of images, it would only succeed in rendering

images that have close views to the training images, while having a severely deteriorating performance for rendering the images that have significantly deviating views from the training images. This issue motivates us to introduce the affine view augmentation module, which aims to augment the number of training images via an introduced differentiable affine warping. By effectively expanding the range of viewpoint variations seen during training, this module helps alleviate overfitting to sparse viewpoints and improve the quality of novel-view satellite image synthesis.

In sum, our main contributions include:

- We explore the affine version of 3DGS in the designed affine 3D Gaussian Splatting module, where a new type of affine projection estimation is explored to estimate the affine projection parameters directly from the training data. The explored affine projection estimation approach could avoid the accumulated error problem that has to be suffered by its RPC-model-based counterpart EOGS (Aira et al., 2025). Moreover, unlike the original 3DGS that is available for handling plane-array images, the explored affine 3DGS could handle line-array satellite images effectively, which is expected to provide a general rendering technique for follow-up works in the satellite image rendering task.
- We design the seasonal embedding module by introducing a season embedding into each 3D Gaussian representation. This module could empower each 3D Gaussian representation to tolerate textural variations among the satellite images captured in different seasons.
- We design the affine view augmentation module, which could synthesize satellite images from different views as additional training supervision. By integrating this module with the aforementioned two modules, the SA-GS is proposed for satellite image rendering. Its effectiveness has been verified by the results in Section 5.

The rest of this paper is organized as follows: Section 2 presents a review on the related works. Section 3 introduces the proposed SA-GS in detail. Section 4 describes experimental details. Section 5 reports experimental results. Section 6 concludes the paper.

2. Related work

In this section, we review the related works on Neural Radiance Fields (NeRF) and 3D Gaussian Splatting (3DGS) applied to satellite imagery.

2.1. Neural radiance fields for satellite imagery

In recent years, Neural Radiance Fields (NeRF) (Mildenhall et al., 2021) have emerged as a popular technique for novel view rendering, and a number of works have extended this technique to satellite images. Satellite images are generally captured by line-array sensors. The imaging mechanism of a line-array sensor is that the sensor captures a line of imagery (i.e., a 1-dimensional image) at a time, and accordingly as the sensor moves, 1-dimensional images are captured at regular time intervals, then the ensemble of these 1-dimensional images constitutes a 2-dimensional image. The above imaging mechanism of line-array sensors is significantly different from the imaging mechanism of plane-array sensors whose imaging model could be approximately considered as the pinhole camera model. This is to say, line-array sensors are not subject to the pinhole camera model. However, it is noted that most of the existing ray-casting-based methods in literature were explored under the pinhole camera model; hence, their performances would be degraded inevitably when they handled the images captured by line-array sensors without additional operations. This issue encourages researchers to investigate specially designed line-array-sensor-based ray sampling strategies for ray-casting-based rendering.

To handle unconstrained photo collections with significant appearance variations, Martin-Brualla et al. (2021) extended the NeRF technique by introducing per-image appearance and transient embeddings, enabling the model to account for variable illumination and occlusions. Derksen and Izzo (2021) proposed S-NeRF, which further adapted the NeRF technique to satellite images by explicitly modeling solar and sky illumination, improving the appearance rendering in regions with shadows. Built on S-NeRF, Marí et al. (2022) incorporated RPC models and a shadow-aware irradiance module with uncertainty weighting, improving the robustness to illumination changes and transient objects. Marí et al. (2023) further improved the altitude estimation accuracy by introducing a shadow rendering process consistent with the scene geometry and incorporating learnable parameters to handle raw images without pre-processing. Gableman and Kak (2024) introduced season-aware rendering by incorporating time-of-year as an additional input, enabling the rendering of seasonal features independent of viewing and solar angles.

Several NeRF-based methods have been proposed to estimate digital surface models (DSMs) for recovering 3D scene geometry from satellite imagery. Qu and Deng (2023) proposed a one-stage implicit surface reconstruction framework that generated dense DSMs and 3D mesh models from multi-view satellite images by learning a signed distance function (SDF) through volume rendering, while incorporating multi-view stereo constraints to mitigate illumination changes and seasonal variations. Wan et al. (2024) introduced a geometry-constrained NeRF-based satellite photogrammetry method, in which an occupancy-guided ray sampling strategy was employed to explicitly regularize the reconstructed geometry and improve DSM quality. More recently, Liu et al. (2025) proposed a depth and normal-guided neural surface reconstruction method, where progressively introduced geometric priors further enhanced the accuracy and stability of the estimated DSMs.

Moreover, some works focused on mitigating the high computational cost and slow rendering speed of NeRF-based methods for satellite imagery. Billouard et al. (2024) extended the Instant-NGP framework (Müller et al., 2022) for multi-date satellite imagery by adopting an efficient sampling strategy together with hash-based feature encoding, achieving faster training and rendering. T. Zhang et al. (2024) proposed a tensor-based scene representation that replaced conventional MLPs with a multiscale tensor decomposition, significantly improving computational efficiency. X. Zhou et al. (2024) further enhanced the efficiency by combining hash-based representations with an improved irradiance modeling strategy to better handle multi-season color and texture variations. It is worth noting that although these methods have achieved acceleration through more compact scene representations or simplified network architectures, their rendering pipelines still rely on the ray-casting-based volume rendering, where the rendering efficiency is inherently constrained by per-ray sampling and accumulation.

2.2. 3D Gaussian splatting for satellite imagery

3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) has shown competitive rendering performances with NeRF on plane-array images, while achieving a much faster speed. During the conventional 3DGS process, an existing SfM (Structure-from-Motion) method (e.g., an incremental SfM method (Schonberger and Frahm, 2016; Cui et al., 2019; Xu et al., 2021) that registers cameras sequentially, or a global SfM method (Dong et al., 2020, 2024; Liu and Dong, 2025) that estimates all camera poses simultaneously) is used to calculate the sparse point cloud and optimize both the intrinsic and extrinsic camera parameters at first. Accordingly, the obtained 3D points are used to initialize a set of 3D Gaussian representations. Then, the set of Gaussian representations is optimized by minimizing the reprojection errors during rendering.

Some preliminary 3DGS-based methods for satellite image rendering straightforwardly employed the conventional 3DGS technique or its variants to handle environmental variations. For example, Dahmani et al. (2024) introduced transient Gaussians into 3DGS to handle transient objects in an unsupervised manner by modeling appearances that captured photometric variations. Xu et al. (2024) replaced the original image-space feature modeling in 3DGS with explicit alignment of pixel appearances to local Gaussians by triplane sampling, enabling better rendering of high-frequency details. D. Zhang et al. (2024) introduced an intrinsic-dynamic disentangled Gaussian appearance model with adaptive sampling and visibility maps for more accurate rendering. Kulhanek et al. (2024) further addressed varying illumination by leveraging robust DINO features (Oquab et al., 2023) with an appearance-aware Gaussian modeling module.

Some recent methods focused on introducing additional constraints or operations to handle seasonal and shadow changes. Wang et al. (2024) proposed the PR3DGS, which integrated a pseudo-view regularization module that leveraged CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) to distinguish rendered images from different views for few-shot novel view synthesis. Bai et al. (2025) proposed an appearance-adaptive Gaussian Splatting method, which could adjust for seasonal features and shadows based on image appearances and solar angles for better rendering quality.

It is noted that since line-array satellite images are not subject to the pinhole camera model, the rendering performances of the above methods on line-array satellite images are still limited. Unlike them, an affine version of 3DGS is explored under the affine camera model in this paper, which could effectively handle line-array satellite images in principle. Moreover, the proposed SA-GS integrates the affine 3DGS with both the seasonal embedding and the affine view augmentation modules, which could adapt to satellite images captured in different seasons while improving its performance.

3. Methodology

In this section, the proposed SA-GS is introduced in detail. The architecture of SA-GS is described first. Then, the three main modules in SA-GS are introduced respectively. Finally, the loss function is given.

3.1. Architecture

The architecture of the proposed SA-GS is shown in Fig. 2. SA-GS takes small-sized images (that are obtained by partitioning the large-sized satellite images) as input, consisting of an affine 3D Gaussian Splatting module, a seasonal embedding module, and an affine view augmentation module. The affine 3D Gaussian Splatting module is explored as the rendering pipeline for building a set of 3D Gaussians. The seasonal embedding module is designed to inject a season embedding into each Gaussian representation for improving its seasonal discrimination capability, while the affine view augmentation module is designed to generate synthetic satellite images as additional training supervision.

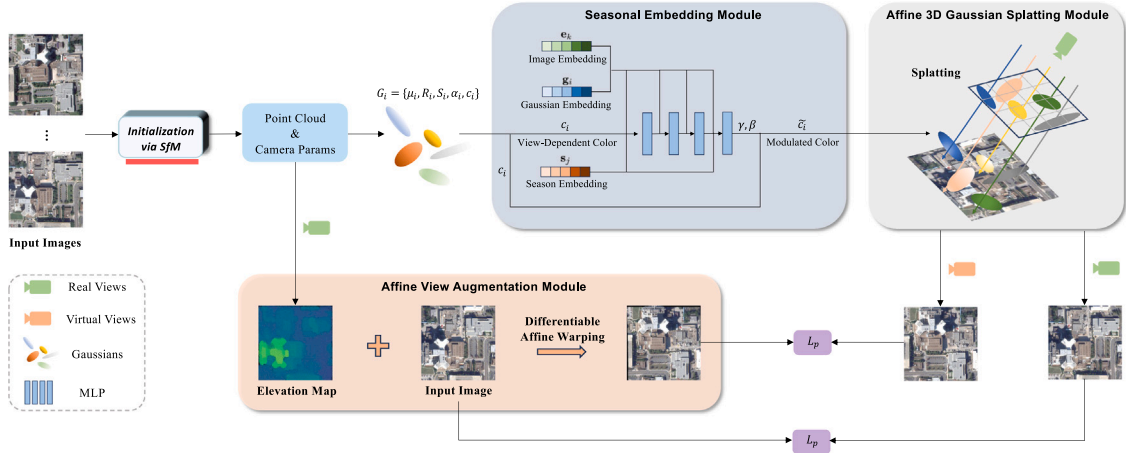


Fig. 2. Architecture of SA-GS. SA-GS consists of an affine 3D Gaussian Splatting module, a seasonal embedding module, and an affine view augmentation module. During training, the input images are processed by the existing SfM method (Wang et al., 2022) to estimate the affine camera parameters and reconstruct 3D scene points, which are used as initials of 3D Gaussians. These Gaussian initials are injected with additional embeddings and passed into the seasonal embedding module for extended representations. Then, the extended Gaussian representations are iteratively optimized by implementing the affine 3DGS explored in the affine 3D Gaussian Splatting module, under the supervision of both the input images and the synthetic images generated by the affine view augmentation module.

The training stage consists of an initialization step and a 3D Gaussian learning step: At the initialization step, the existing SfM method (Wang et al., 2022) is employed to estimate the affine camera parameters and reconstruct 3D scene points from the input small-sized images. Accordingly, the obtained 3D scene points are used as initials of 3D Gaussians. At the 3D Gaussian learning step, the seasonal embedding module takes both the Gaussian initials and the additional embeddings as its input, and outputs extended Gaussian representations. Then under the supervision of both the input real images and the synthetic images generated by the affine view augmentation module, the extended Gaussian representations are iteratively optimized by implementing the explored affine version of 3D Gaussian Splatting in the affine 3D Gaussian Splatting module.

At the inference stage, as done in Kulhanek et al. (2024), only the image embedding of the test image is optimized, while all other model parameters remain fixed. After optimization, the final Gaussian representations are obtained through the seasonal embedding module, and the rendered image is generated through the affine 3D Gaussian Splatting module by implementing the affine 3DGS.

In the following subsections, we will introduce the three modules of SA-GS respectively in detail, including the affine 3D Gaussian Splatting module, the seasonal embedding module, and the affine view augmentation module.

3.2. Affine 3D Gaussian splatting module

The affine 3D Gaussian Splatting module is designed to render line-array satellite images under the affine camera model, where an affine version of 3D Gaussian Splatting is explored. This module takes the 3D Gaussians as input, and outputs the rendered image under the given affine camera parameters. We first provide a brief review on the conventional 3DGS technique, followed by the introduction of the affine 3DGS explored in this module.

Conventional 3DGS: 3DGS (Kerbl et al., 2023) explicitly models scenes using a set of anisotropic 3D Gaussian ellipsoids. Each 3D Gaussian, G_i , is characterized by the 3D center position $\mu_i \in \mathbb{R}^3$ and the 3D covariance matrix $\Sigma_i \in \mathbb{R}^{3 \times 3}$, which describe its properties in the world coordinate system as follows:

$$G_i(x) = e^{-\frac{1}{2}(x-\mu_i)^T \Sigma_i^{-1}(x-\mu_i)}, \quad (1)$$

where Σ_i is typically parameterized by a scaling matrix $S_i \in \mathbb{R}^3$ and a rotation matrix $R_i \in SO(3)$:

$$\Sigma_i = R_i S_i S_i^T R_i^T. \quad (2)$$

Beyond μ_i and Σ_i , each 3D Gaussian also includes the opacity $\alpha_i \in (0, 1)$ for controlling its transparency and the Spherical Harmonic (SH) coefficients $sh_i \in \mathbb{R}^H$ (H denotes the degree of SH coefficients) for representing its view-dependent color.

For rendering, 3D Gaussians are projected onto the 2D image plane, a process originally developed for the pinhole camera model. Given a viewing transformation matrix W and the Jacobian matrix J , which is the affine approximation of the projective transformation, the 2D covariance matrix Σ'_i and the 2D center position μ'_i are formed as:

$$\Sigma'_i = J W \Sigma_i W^T J^T, \quad \mu'_i = J W \mu_i. \quad (3)$$

These projected 2D Gaussians are then α -blended to compute the final color $C(p)$ for each pixel p as follows:

$$C(p) = \sum_{i \in \mathcal{N}} c_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j), \quad (4)$$

where $\mathcal{N}$ is the set of Gaussians overlapping pixel p , c_i is the view-dependent color of the i th Gaussian, and α_i is its screen-space opacity, which is defined as:

$$\alpha_i = \alpha_i e^{-\frac{1}{2}(y-\mu'_i)^T \Sigma_i'^{-1}(y-\mu'_i)}, \quad (5)$$

where y represents the Gaussian's coordinates in the 2D space.

Affine 3DGS: In contrast to the conventional 3DGS, the affine version of 3D Gaussian Splatting is explored under the affine camera model that directly performs a linear projection from 3D world coordinates to 2D image coordinates. Accordingly, the projection of the Gaussian's center μ_i and covariance Σ_i defined in Eq. (3) is reformulated to:

$$\Sigma'_i = M \Sigma_i M^T, \quad \mu'_i = M \mu_i + t, \quad (6)$$

where $M \in \mathbb{R}^{2 \times 3}$ is the linear mapping matrix and $t \in \mathbb{R}^{2 \times 1}$ is the translation vector of the affine camera model defined as $A = [M|t]$. This projection formulation enables the rendering of line-array satellite images under the affine camera model.

3.3. Seasonal embedding module

As shown in Fig. 2, the seasonal embedding module takes both the 3D Gaussians and the additional embeddings as input, and outputs extended Gaussian representations with modulated colors to improve the seasonal discrimination capability of each 3D Gaussian representation, where the color modulation of each 3D Gaussian is realized via a proposed dual-embedding MLP.

In this module, three types of learnable embeddings are introduced to capture appearance information at different levels, including the image embeddings $\{e_k\}_{k=1}^K$, the Gaussian embeddings $\{g_i\}_{i=1}^N$, and the season embeddings $\{s_j\}_{j=1}^S$, where K , N , and S denote the number of input images, Gaussians, and seasons respectively. The image embedding e_k captures global illumination and appearance characteristics specific to each input image, while the Gaussian embedding g_i models local variations across different regions of the scene. Then, considering that seasonal variations could lead to color shifts of scenes, the season embedding s_j is introduced to modulate the color of each Gaussian representation. Specifically, for each Gaussian, a linear transformation is applied, as this transformation could accommodate a wide range of color changes. The modulated color $\tilde{c}_i$ is then computed as:

$$\tilde{c}_i = \gamma \cdot c_i + \beta, \quad (7)$$

where c_i denotes the Gaussian's view-dependent color, and γ and β are the linear transformation parameters outputted by an MLP f with four hidden layers:

$$(\gamma, \beta) = f(e_k, g_i, s_j, c_i), \quad (8)$$

where the image embedding e_k , the Gaussian embedding g_i , the season embedding s_j , and the Gaussian's view-dependent color c_i serve as the MLP's input. The extended Gaussian representations with the modulated colors are subsequently passed into the affine 3D Gaussian Splatting module for final rendering, and the three types of embeddings in the seasonal embedding module are jointly optimized with the 3D Gaussians during training.

Here, we employ a dual-embedding MLP, in which both the image embedding e_k and the Gaussian embedding g_i are incorporated into every layer. This design ensures that each layer is aware of both global and local appearance cues, enhancing the network's capability to model fine appearance variations. Meanwhile, the season embedding s_j is introduced only at the first and the final layer, enabling the MLP to disentangle season-specific information from general appearance modeling.

3.4. Affine view augmentation module

Considering that the number of satellite images for a given scene is generally small and the model is prone to overfitting on such sparse training views, the affine view augmentation module is designed to generate synthetic satellite images as additional training supervision. This module takes the input image and the corresponding elevation map projected from the reconstructed SfM point cloud as input, and outputs the synthetic satellite image under the given view via the differentiable affine warping, as shown in Fig. 2.

Here, the differentiable affine warping under the affine camera model is explored first. Specifically, the forward projection from a 3D point P in the world space to its corresponding pixel p under the affine camera model A is formulated as:

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} & t_1 \\ m_{21} & m_{22} & m_{23} & t_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = A \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}, \quad (9)$$

where (u, v) denote the image coordinates of the pixel p , and (X, Y, Z) denote the world coordinates of the 3D point P . Then, given the elevation value Z , the inverse projection from pixel p back to the corresponding 3D point P is formulated as:

$$\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} M_{11} & M_{12} & M_{13} & M_{14} \\ M_{21} & M_{22} & M_{23} & M_{24} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & Z \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \quad (10)$$

$$= A^{\text{inv}}(Z) \begin{bmatrix} u \\ v \\ 1 \end{bmatrix},$$

where the coefficients $\{M_{ij}\}$ are the rational functions of the affine camera parameters $\{m_{ij}\}$ and $\{t_i\}$, defined as:

$$\begin{cases} M_{11} = \frac{-m_{22}}{m_{12}m_{21} - m_{11}m_{22}}, & M_{21} = \frac{m_{21}}{m_{12}m_{21} - m_{11}m_{22}}, \\ M_{12} = \frac{m_{12}}{m_{12}m_{21} - m_{11}m_{22}}, & M_{22} = \frac{-m_{11}}{m_{12}m_{21} - m_{11}m_{22}}, \\ M_{13} = \frac{m_{13}m_{22} - m_{12}m_{23}}{m_{12}m_{21} - m_{11}m_{22}}, & M_{23} = \frac{m_{11}m_{23} - m_{13}m_{21}}{m_{12}m_{21} - m_{11}m_{22}}, \\ M_{14} = \frac{t_1m_{22} - m_{12}t_2}{m_{12}m_{21} - m_{11}m_{22}}, & M_{24} = \frac{m_{11}t_2 - t_1m_{21}}{m_{12}m_{21} - m_{11}m_{22}}. \end{cases} \quad (11)$$

According to Eqs. (9) and (10), the mapping relation of the differentiable affine warping is defined as:

$$p_1 = A_1 A_0^{\text{inv}}(Z_0) p_0, \quad (12)$$

where p_1 denotes the target pixel aligned with the source pixel p_0 at elevation Z_0 , and A_1 and A_0^{inv} denote the corresponding affine camera model and the inverse projection matrix respectively.

Based on the explored differentiable affine warping, the affine view augmentation module first selects a source image I_{src} from the real views, and then generates the target image I_{tgt} corresponding to a virtual view. To generate such virtual views, we perturb the viewing direction of the affine camera corresponding to the real view with the Gaussian-distributed noise, thus creating a set of virtual affine cameras.

Specifically, using the forward projection defined in Eq. (9), the reconstructed SfM point cloud is projected onto the image plane under the real view, obtaining the elevation value Z_{src} for each pixel p_{src} in I_{src} . With this elevation, each pixel p_{src} is then warped to its corresponding target pixel p_{tgt} under the virtual via the differentiable affine warping defined in Eq. (12), which is formulated as:

$$p_{\text{tgt}} = A_{\text{vir}} A_{\text{real}}^{\text{inv}}(Z_{\text{src}}) p_{\text{src}}, \quad (13)$$

where A_{vir} denotes the affine camera model of the virtual view, and $A_{\text{real}}^{\text{inv}}$ denotes the inverse projection matrix of the real view. To synthesize the target image of the virtual view, the color value at the projected pixel p_{tgt} can be obtained through $I_{\text{src}}(p_{\text{src}})$. However, since p_{tgt} is typically non-integer, a reversed bilinear sampling strategy (Wang et al., 2018) is employed to distribute its color to the four nearest pixel neighbors $\{(u_i, v_i)\}_{i=0}^3$, and the interpolation weight assigned to each of them is defined as:

$$w(u_i, v_i) = (1 - |u_i - u_{\text{tgt}}|)(1 - |v_i - v_{\text{tgt}}|), \quad (14)$$

where $(u_{\text{tgt}}, v_{\text{tgt}})$ denote the projected coordinates of p_{tgt} in the target image. Finally, by aggregating the contributions from the corresponding pixels, the synthesized color of each pixel (u, v) in the target image I_{tgt} of the virtual view is computed as:

$$I_{\text{tgt}}(u, v) = \frac{\sum_{p_{\text{src}} \in \Omega} w(u, v) \cdot I_{\text{src}}(p_{\text{src}})}{\sum_{p_{\text{src}} \in \Omega} w(u, v)}, \quad (15)$$

where Ω denotes the set of all pixels in I_{src} that are warped to the pixel (u, v) in the target image. These warped images are used as additional supervision for training SA-GS.

3.5. Loss function

The total loss function of the proposed SA-GS contains three terms: a photometric loss L_p , an uncertainty loss L_{uc} , and a perceptual loss L_{lips} .

Following Kerbl et al. (2023), the photometric loss is defined as a weighted sum of the L_1 loss and the SSIM loss (Wang et al., 2004), which measures the pixel-level errors between the rendered and reference images:

$$L_p = \lambda_p L_1(M \odot \tilde{I}, M \odot I) + (1 - \lambda_p)(1 - \text{SSIM}(M \odot \tilde{I}, M \odot I)), \quad (16)$$

where I is the reference image, $\tilde{I}$ is the rendered image with Gaussians' modulated colors, $\hat{I}$ is the rendered image with Gaussians' view-dependent colors, λ_p is a tuning parameter, “ $\odot$ ” is the Hadamard product, and M is a visibility mask introduced in Kulhanek et al. (2024), defined as:

$$M = \min \left(1, \frac{1}{2\sigma^2} \right), \quad (17)$$

where σ is the uncertainty map predicted by an uncertainty predictor as implemented in Kulhanek et al. (2024).

To jointly optimize the uncertainty map σ during the 3DGS optimization, the uncertainty loss L_{uc} from Kulhanek et al. (2024) is used:

$$L_{uc} = \frac{\min \left(1, 2 - \frac{2\tilde{D} \cdot D}{\|\tilde{D}\|_2 \|D\|_2} \right)}{2\sigma^2} + \lambda_\sigma \log \sigma, \quad (18)$$

where $\tilde{D}$ is the DINO features (Oquab et al., 2023) of the rendered image, D is the DINO features of the reference image, and λ_σ is a tuning parameter. Here, DINO features are used following WildGaussians (Kulhanek et al., 2024) to measure the feature-level similarity of different images, which are extracted from large-scale pretrained vision transformers and could capture high-level semantic and structural information of images. Due to their strong representation capability, DINO features have been widely adopted in various visual tasks (Yang et al., 2024; S. Zhou et al., 2024; Liu et al., 2024). Because of their robustness to appearance variations, including illumination changes and seasonal differences, DINO features provide a more reliable similarity measure for uncertainty in our method, which enables the uncertainty predictor to distinguish appearance-induced discrepancies from true reconstruction errors, and thus produce more reliable uncertainty estimation during optimization.

In addition, the perceptual loss L_{lpipe} (Zhang et al., 2018) is used to improve the perceptual quality of the rendered images. Accordingly, the total loss function L is formulated as:

$$L = L_p + \lambda_1 L_{uc} + \lambda_2 L_{lpipe}, \quad (19)$$

where λ_1 and λ_2 are tuning parameters.

Note: The degree of Spherical Harmonic coefficients is an important parameter in both the standard 3DGS and the proposed affine 3DGS. According to the definition of SH (Kerbl et al., 2023), the degree of SH coefficients directly influences the modeling ability of 3DGS. Theoretically, the higher the degree of SH coefficients is, the stronger the modeling ability of the 3DGS is. This issue means that the setting of the degree of SH coefficients would inevitably influence the performance of the proposed affine 3DGS, whose seasonal embedding module also uses the SH coefficients as part of its MLP input. In Section 5.2.3, we would investigate such an influence by evaluating the proposed method under different settings of the degree of SH coefficients.

4. Experiments

4.1. Dataset and evaluation metrics

Dataset. We conduct experiments on the 2019 IEEE GRSS Data Fusion Contest (DFC2019) dataset (Le Saux et al., 2019), which provides high-resolution images captured by Maxar WorldView-3 between 2014 and 2016 over the cities of Jacksonville and Omaha in the USA. Unlike the existing works (Derksen and Izzo, 2021; Marí et al., 2022) where only 4 areas of interest (AOIs) from Jacksonville are used for evaluation, we use 8 AOIs from this dataset in this paper, including 4 AOIs from Jacksonville and 4 AOIs from Omaha. For each AOI, the ground truth digital surface model (DSM) is provided, covering an area of 256 by 256 m. Following Derksen and Izzo (2021) and Marí et al. (2022), the original images are cropped to approximately 800 by 800 pixels according to the DSM coverage, with a ground sample distance (GSD) of 0.3 m per pixel.

Table 1

Index and training/test split for AOIs from Jacksonville and Omaha.

Jacksonville				
AOI index	JAX 004	JAX 068	JAX 214	JAX 260
Training/Test	7/4	15/4	20/4	13/4
Omaha				
AOI index	OMA 132	OMA 163	OMA 203	OMA 281
Training/Test	35/6	34/6	37/6	35/6

Details of the AOI index and training/test image split used in our work are shown in Table 1, where the split ensures that test images cover different seasons to better evaluate each method's performance under seasonal and environmental variations.

Evaluation metrics. As done in Mildenhall et al. (2021) and Kerbl et al. (2023), the following evaluation metrics are used in our work to evaluate the quality of rendered images: the structural similarity index (SSIM) (Wang et al., 2004), the peak signal-to-noise ratio (PSNR), and the learned perceptual image patch similarity (LPIPS) (Zhang et al., 2018).

4.2. Implementation details

The code of the proposed SA-GS is written and implemented in PyTorch, where CUDA kernels are used to handle the rasterizer for the affine 3D Gaussian Splatting. SA-GS is trained on a single NVIDIA RTX 3090 Ti GPU using the Adam optimizer. At the training stage, the model is trained for 20,000 iterations for each AOI, and Gaussian densification and pruning are performed every 100 iterations between iterations 500 and 10,000. In the affine 3D Gaussian Splatting module, the degree of SH coefficients H is set to 3, and the background color for rendering is black. In the seasonal embedding module, the same degree of SH coefficients is used for the MLP input, and the index of the season embedding for each input image is determined by its acquisition time. In the affine view augmentation module, 200 virtual views are generated for each AOI. The opacity decay strategy from Han et al. (2024) is adopted to filter out the Gaussians that indicate incorrect geometry. As done in Kulhanek et al. (2024), the parameters for the loss functions are set to $\lambda_p = 0.8$ and $\lambda_\sigma = 0.5$. The tuning parameters in Eq. (19) are set to $\lambda_1 = 1.0$ and $\lambda_2 = 0.005$. At the inference stage, following Kulhanek et al. (2024), the image embedding of the test image is optimized for 128 iterations.

4.3. Comparative methods

We evaluate the proposed SA-GS in comparison to 11 state-of-the-art methods for the satellite image rendering task, including 5 NeRF-based methods (S-NeRF (Derksen and Izzo, 2021), Sat-NeRF (Marí et al., 2022), Season-NeRF (Gableman and Kak, 2024), Instant-NGP (Müller et al., 2022), and SAT-NGP (Billouard et al., 2024)) and 6 3DGS-based methods (3DGS (Kerbl et al., 2023), Mip-Splatting (Yu et al., 2024), GS-W (D. Zhang et al., 2024), WildGaussians (Kulhanek et al., 2024), EOGS (Aira et al., 2025), and SatGS (Bai et al., 2025)). It has to be explained that the best versions of S-NeRF and Sat-NeRF are used for comparison. For the comparative 3DGS-based methods, EOGS (Aira et al., 2025) is initialized following its official implementation, while other methods are initialized following (Bai et al., 2025). The results of all comparative methods are either obtained from the authors' released codes or reproduced by us under the same experimental settings.

5. Results

5.1. Comparative evaluation

Here, we evaluate the proposed SA-GS and 11 comparative state-of-the-art methods on 8 AOIs from the DFC2019 dataset (Le Saux et al., 2019).

Table 2

Comparison on four AOIs from Jacksonville (AOIs “JAX 004”, “JAX 068”, “JAX 214”, and “JAX 260”). ↓/↑ denotes that lower/higher is better. The best results are in **bold** in each metric.

Method	JAX 004			JAX 068			JAX 214			JAX 260		
	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓
S-NeRF	0.642	22.54	0.391	0.794	24.32	0.262	0.689	20.94	0.368	0.611	21.15	0.408
Sat-NeRF	0.646	22.93	0.382	0.822	24.89	0.242	0.720	21.08	0.345	0.660	21.64	0.385
Season-NeRF	0.616	21.72	0.441	0.706	21.31	0.304	0.673	20.55	0.409	0.606	20.51	0.469
Instant-NGP	0.605	21.14	0.436	0.707	22.53	0.364	0.649	20.33	0.422	0.574	21.08	0.469
SAT-NGP	0.613	21.27	0.455	0.715	23.09	0.358	0.653	20.76	0.419	0.578	20.77	0.486
3DGS	0.441	18.21	0.487	0.711	19.66	0.268	0.584	16.85	0.368	0.445	17.57	0.457
Mip-Splatting	0.451	19.21	0.478	0.714	19.75	0.260	0.601	17.12	0.352	0.499	17.95	0.447
GS-W	0.533	19.92	0.425	0.808	23.33	0.195	0.693	20.22	0.299	0.631	20.42	0.381
WildGaussians	0.564	22.32	0.404	0.802	24.20	0.209	0.679	20.78	0.345	0.638	22.38	0.394
EOGS	0.640	22.23	0.402	0.771	19.77	0.253	0.633	16.84	0.353	0.620	19.59	0.417
SatGS	0.619	23.28	0.365	0.819	24.57	0.193	0.697	20.79	0.313	0.673	23.28	0.362
SA-GS (ours)	0.653	23.86	0.346	0.850	25.59	0.165	0.733	22.09	0.293	0.683	23.54	0.356

Table 3

Comparison on four AOIs from Omaha (AOIs “OMA 132”, “OMA 163”, “OMA 203”, and “OMA 281”).

Method	OMA 132			OMA 163			OMA 203			OMA 281		
	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓
S-NeRF	0.595	18.25	0.404	0.532	17.92	0.446	0.645	18.77	0.417	0.607	19.23	0.454
Sat-NeRF	0.610	18.94	0.397	0.555	18.49	0.443	0.604	18.99	0.517	0.630	19.81	0.438
Season-NeRF	0.611	20.29	0.436	0.528	19.00	0.491	0.639	20.15	0.440	0.578	18.89	0.504
Instant-NGP	0.559	17.77	0.475	0.468	17.22	0.516	0.594	18.20	0.472	0.570	17.73	0.488
SAT-NGP	0.557	18.24	0.478	0.472	17.19	0.517	0.595	17.58	0.484	0.562	18.67	0.490
3DGS	0.380	16.46	0.515	0.392	16.36	0.496	0.502	17.33	0.486	0.412	16.20	0.513
Mip-Splatting	0.529	18.36	0.409	0.507	17.58	0.418	0.632	18.82	0.354	0.592	17.81	0.397
GS-W	0.551	18.65	0.392	0.544	18.39	0.386	0.651	19.66	0.338	0.621	18.87	0.367
WildGaussians	0.600	20.14	0.381	0.533	18.85	0.424	0.595	17.58	0.421	0.623	19.72	0.398
EOGS	0.550	18.05	0.548	0.452	16.63	0.570	0.622	18.26	0.482	0.602	17.16	0.465
SatGS	0.601	20.62	0.361	0.530	18.88	0.401	0.592	18.10	0.390	0.614	19.56	0.382
SA-GS (ours)	0.685	23.11	0.310	0.625	20.73	0.357	0.699	22.36	0.319	0.688	21.59	0.343

Table 4

Comparison of Sat-NeRF (Marí et al., 2022), SatGS (Bai et al., 2025), and the proposed SA-GS on AOIs “JAX 068 (Fall)” and “OMA 281 (Spring)”.

Method	JAX 068 (Fall)			OMA 281 (Spring)		
	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓
Sat-NeRF	0.683	21.69	0.349	0.737	22.63	0.340
SatGS	0.696	21.32	0.278	0.672	22.17	0.349
SA-GS	0.788	23.52	0.203	0.749	23.29	0.292

5.1.1. Results on Jacksonville

We firstly conduct the experiments on 4 AOIs from Jacksonville, and the corresponding results are reported in Table 2. Three points could be observed from this table: (i) The conventional 3DGS (Kerbl et al., 2023) and Mip-Splatting (Yu et al., 2024) exhibit poor performances in most cases. (ii) By introducing additional constraints or operations to handle environmental variations, some 3DGS variants (D. Zhang et al., 2024; Kulhanek et al., 2024; Aira et al., 2025; Bai et al., 2025) achieve relatively better performances. However, the improvements are still limited, and in some cases, they even perform worse than the NeRF-based methods (Derksen and Izzo, 2021; Marí et al., 2022; Gableman and Kak, 2024; Müller et al., 2022; Billouard et al., 2024). (iii) The proposed SA-GS achieves the best performance among all the comparative methods under all AOIs and metrics, demonstrating its effectiveness in handling line-array satellite images.

In addition, Fig. 3 shows four visualization examples of the proposed SA-GS as well as six comparative methods that perform relatively better than the rest of the methods, including S-NeRF (Derksen and Izzo, 2021), Sat-NeRF (Marí et al., 2022), Season-NeRF (Gableman and Kak, 2024), GS-W (D. Zhang et al., 2024), WildGaussians (Kulhanek et al., 2024), and SatGS (Bai et al., 2025). As seen from the yellow rectangles, the comparative NeRF-based methods generally produce blurry and over-smoothed renderings. Additionally, the comparative 3DGS-based

methods suffer from obvious noise and artifacts, particularly in regions with vegetation or shadows. In contrast, the proposed SA-GS not only produces more reliable renderings, but also maintains more delicate object structures with fewer artifacts, as observed in the rendering results of vegetation and houses.

5.1.2. Results on Omaha

We also evaluate the proposed SA-GS on 4 AOIs from Omaha, and the corresponding results are reported in Table 3. As seen from this table, the proposed SA-GS performs significantly better than other comparative methods under all AOIs and metrics, consistent with the results observed on the AOIs from Jacksonville. These results further confirm that SA-GS could effectively handle line-array satellite images even under significant seasonal variations and strong illumination changes.

Additionally, Fig. 4 shows another four visualization examples of the proposed SA-GS as well as six comparative methods, including S-NeRF (Derksen and Izzo, 2021), Sat-NeRF (Marí et al., 2022), Season-NeRF (Gableman and Kak, 2024), GS-W (D. Zhang et al., 2024), WildGaussians (Kulhanek et al., 2024), and SatGS (Bai et al., 2025). As seen from the full images and the yellow rectangles, S-NeRF and Sat-NeRF could not effectively capture seasonal variations, with their rendered images showing incorrect color tones across the entire image due to their inability to associate seasonal changes with local scene structures. Season-NeRF could capture seasonal appearances to some extent, but still fails to preserve fine scene details and clear object boundaries. Additionally, the comparative 3DGS-based methods are also unable to effectively model seasonal appearances, resulting in severe artifacts and noise. In contrast, the proposed SA-GS not only produces accurate global appearances across different seasons, but also better captures local scene changes caused by seasonal variations, such as vegetation growth and snow coverage.

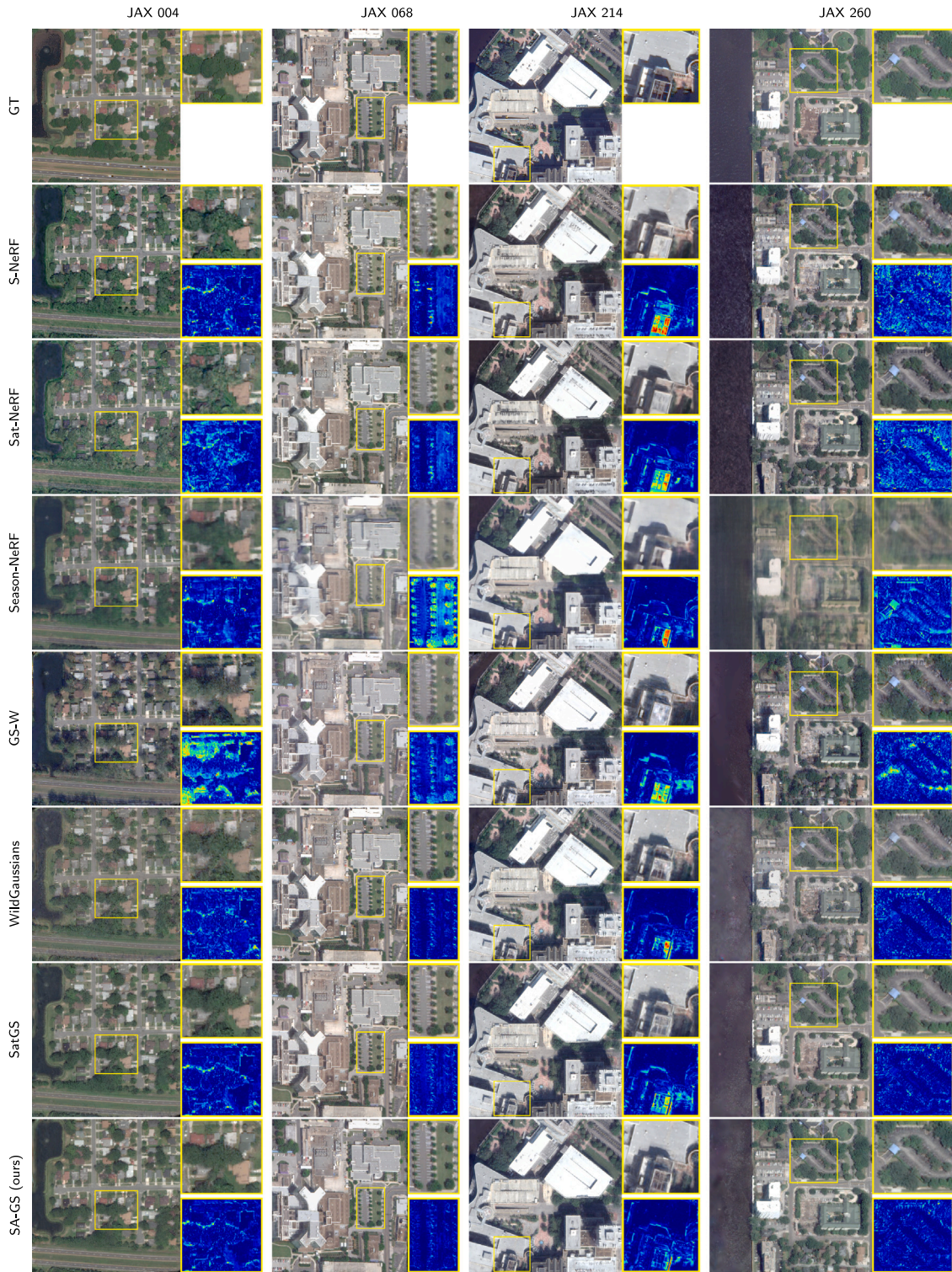


Fig. 3. Visualization results of S-NeRF (Derksen and Izzo, 2021), Sat-NeRF (Marí et al., 2022), Season-NeRF (Gableman and Kak, 2024), GS-W (D. Zhang et al., 2024), WildGaussians (Kulhanek et al., 2024), SatGS (Bai et al., 2025), and our SA-GS on four AOIs from Jacksonville (AOIs “JAX 004”, “JAX 068”, “JAX 214”, and “JAX 260”).

5.1.3. Results on season-dependent datasets

We also conduct the experiments on season-dependent datasets via constructing two season-dependent datasets from the DFC2019 dataset (Le Saux et al., 2019), including a fall-only dataset whose images are all captured in fall for AOI “JAX 068” and a spring-only dataset whose images are all captured in spring for AOI “OMA 281”, denoted as “JAX 068 (Fall)” and “OMA 281 (Spring)” respectively.

We allocate 4 training and 2 test images for “JAX 068 (Fall)”, and 13 training and 2 test images for “OMA 281 (Spring)”. Then, we evaluate the proposed SA-GS in comparison to the NeRF-based method Sat-NeRF (Marí et al., 2022) and the 3DGS-based method SatGS (Bai et al., 2025) that have shown relatively better performances than other methods on the DFC2019 dataset. The corresponding results are reported in Table 4. As seen from this table, the proposed SA-GS still

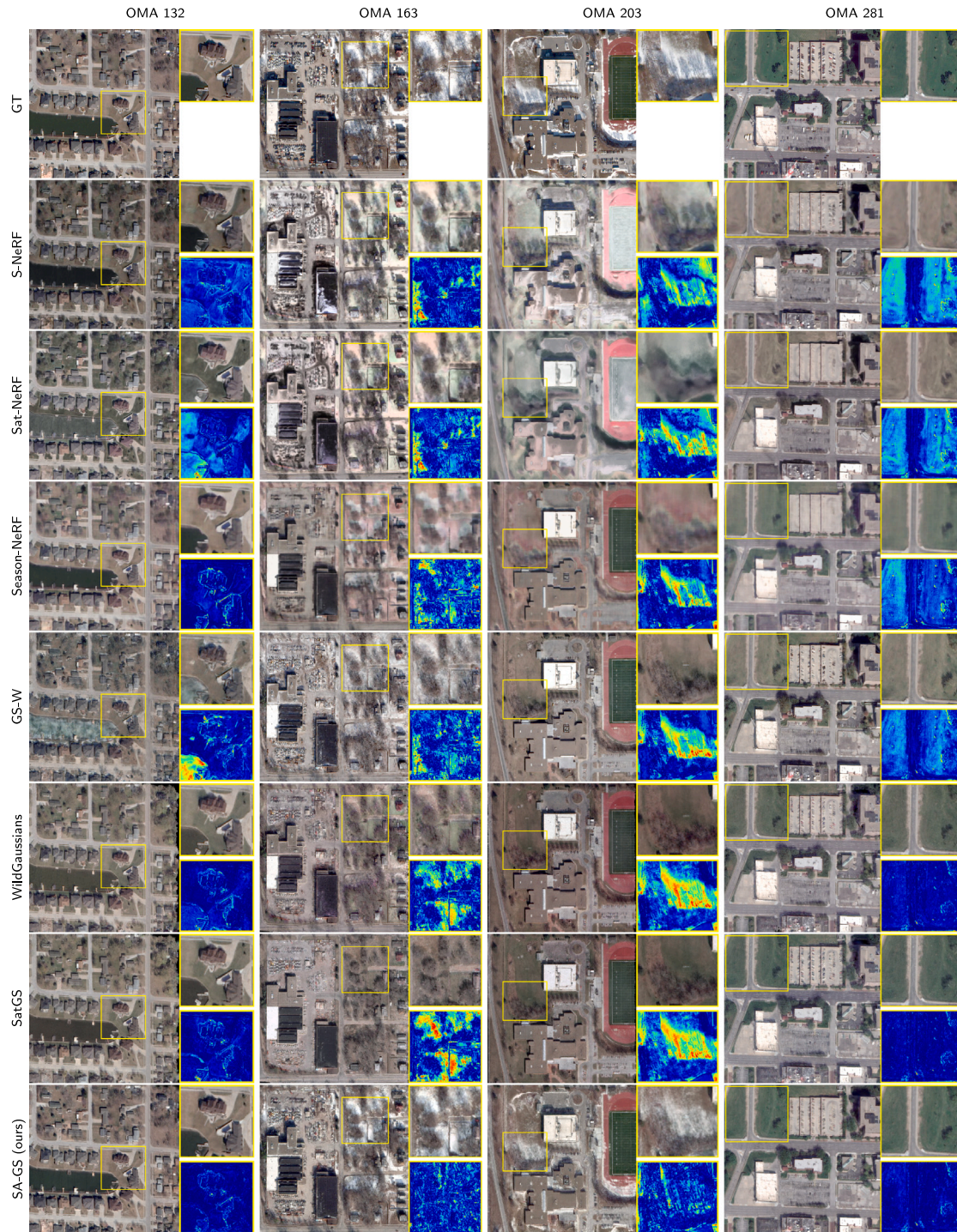


Fig. 4. Visualization results of S-NeRF (Derksen and Izzo, 2021), Sat-NeRF (Marí et al., 2022), Season-NeRF (Gableman and Kak, 2024), GS-W (D. Zhang et al., 2024), WildGaussians (Kulhanek et al., 2024), SatGS (Bai et al., 2025), and our SA-GS on four AOIs from Omaha (AOIs “OMA 132”, “OMA 163”, “OMA 203”, and “OMA 281”).

outperforms the comparative methods, which further demonstrates the effectiveness of the proposed SA-GS even under a single-season setting.

5.2. Ablation studies

To verify the effectiveness of each key element in the proposed SA-GS, we conduct ablation studies on the DFC2019 dataset (Le Saux et al., 2019).

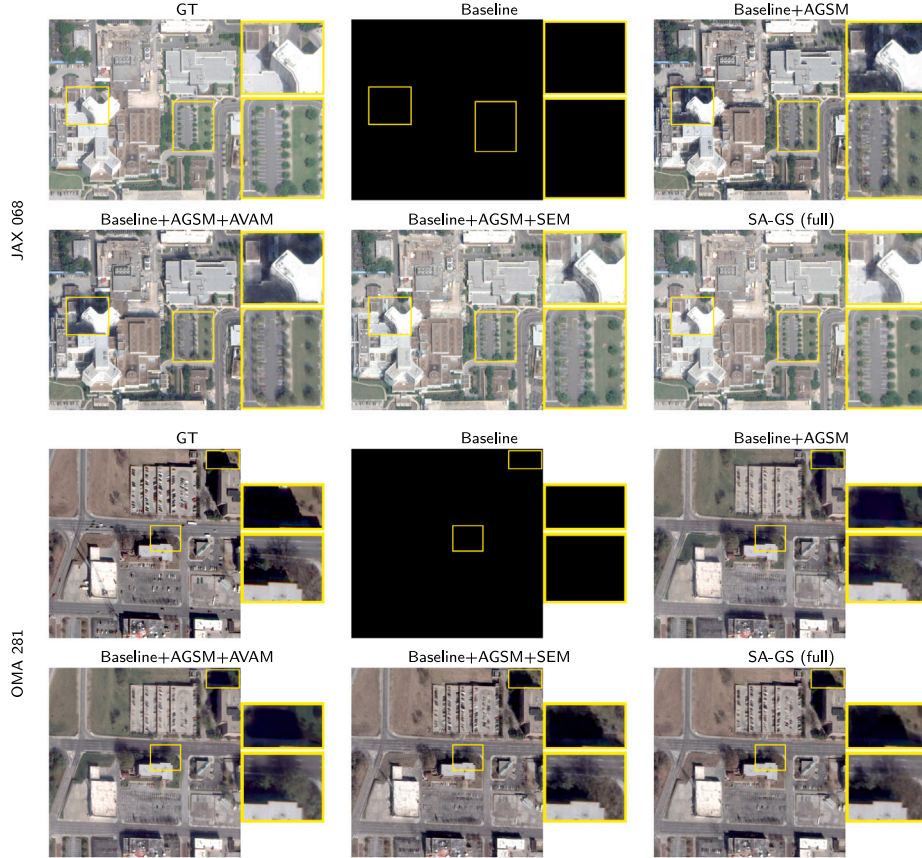
5.2.1. Ablation studies on the main components

The proposed SA-GS consists of three main components, including the affine 3D Gaussian Splatting module (denoted as “AGSM”), the seasonal embedding module (denoted as “SEM”), and the affine view augmentation module (denoted as “AVAM”). Here, the conventional 3DGS (Kerbl et al., 2023) serves as the baseline, and we evaluate SA-GS on AOIs “JAX 068” and “OMA 281” by replacing the rendering pipeline developed under the pinhole camera model with AGSM and adding SEM and AVAM to the baseline respectively. It is noted that under

Table 5

Ablation study for the main components of the proposed SA-GS on AOIs “JAX 068” and “OMA 281”. “AGSM” denotes the affine 3D Gaussian Splatting module, “SEM” denotes the seasonal embedding module, and “AVAM” denotes the affine view augmentation module.

Baseline	AGSM	AVAM	SEM	JAX 068			OMA 281		
				SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓
✓				0.001	4.06	0.724	0.014	5.84	0.700
✓	✓			0.752	20.25	0.234	0.637	18.21	0.370
✓	✓	✓		0.784	20.75	0.216	0.651	18.33	0.368
✓	✓		✓	0.830	25.10	0.179	0.665	21.26	0.354
✓	✓	✓	✓	0.850	25.59	0.165	0.688	21.59	0.343

**Fig. 5.** Visualization results by the main components of the proposed SA-GS on AOIs “JAX 068” and “OMA 281” in the ablation study.**Table 6**

Ablation study for the main components of the proposed seasonal embedding module on AOIs “JAX 068” and “OMA 281”. “SE” denotes the season embeddings, and “DE-MLP” denotes the proposed dual-embedding MLP.

Method	JAX 068			OMA 281		
	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓
SA-GS w/o. SE, DE-MLP	0.842	25.50	0.168	0.676	21.31	0.350
SA-GS w/o. DE-MLP	0.847	25.56	0.166	0.683	21.52	0.345
SA-GS (full)	0.850	25.59	0.165	0.688	21.59	0.343

Table 7

Performance of SA-GS under different settings of the degree of SH coefficients for the MLP input on AOIs “JAX 068” and “OMA 281”.

SH degree	JAX 068			OMA 281		
	SSIM↑	PSNR↑	LPIPS↓	SSIM↑	PSNR↑	LPIPS↓
0	0.845	25.48	0.170	0.682	21.50	0.346
1	0.847	25.52	0.168	0.685	21.55	0.344
2	0.848	25.57	0.166	0.687	21.58	0.343
3	0.850	25.59	0.165	0.688	21.59	0.343

the baseline, the standard pinhole-camera-based rendering pipeline is employed, the original Gaussian representations with view-dependent colors are used for rendering, and only the input real images are used as supervision. The corresponding results are reported in Table 5. As seen from this table, the sequential addition of AGSM, SEM, and AVAM leads to consistent performance improvements, and the full SA-GS achieves the best performance by incorporating all three components, demonstrating their effectiveness.

To further verify the effectiveness of the main components, several visualization results are given in Fig. 5. As shown in the figure, the rendering results of the baseline are entirely black images with all pixels assigned the background color, while their qualities are significantly improved once AGSM is introduced. Furthermore, the proposed SEM improves the model’s capability to discriminate significant textural variations caused by seasonal changes, while the proposed AVAM enhances the reconstruction of more accurate scene details. The full

Table 8

Performance of SA-GS under different values of tuning parameters $\lambda_1 \in \{0, 0.5, 1.0, 2.0\}$ and $\lambda_2 \in \{0, 0.002, 0.005, 0.010\}$ on AOI “JAX 068”.

λ_1	λ_2	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$
0	0	0.848	25.56	0.170
0	0.005	0.849	25.57	0.166
1.0	0	0.848	25.56	0.169
0.5	0.005	0.850	25.58	0.165
2.0	0.005	0.849	25.58	0.165
1.0	0.002	0.849	25.57	0.166
1.0	0.005	0.850	25.59	0.165
1.0	0.010	0.849	25.56	0.164

Table 9

Efficiency comparison of different methods averaged over eight AOIs from the DFC2019 dataset (Le Saux et al., 2019). “SEM” denotes the seasonal embedding module.

Method	Training time (h)	FPS
S-NeRF	25.46	0.043
Sat-NeRF	37.52	0.037
Season-NeRF	4.47	0.005
Instant-NGP	0.96	11.4
SAT-NGP	1.01	11.1
3DGS	0.13	105.9
Mip-Splatting	0.17	102.7
GS-W	1.27	25.1
WildGaussians	1.12	29.6
EOGS	0.09	50.8
SatGS	3.97	8.4
SA-GS w/o. SEM	0.57	96.9
SA-GS	1.18	28.9

SA-GS, which incorporates both components, achieves the best results, as illustrated by the corresponding results in the final column of Fig. 5.

5.2.2. Ablation studies on SEM

To further explore the effectiveness of the proposed seasonal embedding module, we conduct additional ablation studies on AOIs “JAX 068” and “OMA 281”. First, the season embedding (denoted as “SE”) branch is removed, and the proposed dual-embedding MLP (denoted as “DE-MLP”) is replaced with the conventional MLP from Kulhanek et al. (2024), where input embeddings are only processed at the first layer. Then, SE and DE-MLP are sequentially added to the model, and the corresponding results are reported in Table 6. As shown in the table, both SE and DE-MLP improve the model’s performance, with a more significant improvement in AOI “OMA 281” where seasonal variations are stronger, and the full SA-GS achieves the best results under all metrics.

5.2.3. Effect of the SH coefficient degree

In order to evaluate the performance of the proposed method under different settings of the degree of SH coefficients for the MLP input, SA-GS is trained with different degrees in $\{0, 1, 2, 3\}$ on AOIs “JAX 068” and “OMA 281”. The corresponding results are reported in Table 7. As seen from this table, the performance of the proposed method would be improved with the increase of the SH coefficient degree, while the improvement margin would become slight when the degree of SH coefficients is no less than 2. SA-GS with 3rd-degree SH coefficients achieves the best overall performance.

5.2.4. Effect of the tuning parameters λ_1 and λ_2

We also conduct additional ablation studies on AOI “JAX 068” to evaluate the effect of the tuning parameters λ_1 and λ_2 in Eq. (19). Specifically, SA-GS is trained with different values of $\lambda_1 \in \{0, 0.5, 1.0, 2.0\}$ and $\lambda_2 \in \{0, 0.002, 0.005, 0.010\}$, and the corresponding results are reported in Table 8. As seen from the table, when λ_1 ranges from 0 to

2.0, and λ_2 ranges from 0 to 0.010, the corresponding results are close, and SA-GS with $\lambda_1 = 1.0$ and $\lambda_2 = 0.005$ achieves a trade-off among all the evaluation metrics. It demonstrates that the performance of SA-GS is insensitive to the tuning parameters λ_1 and λ_2 .

5.3. Efficiency comparison

Here, we report the average training time in GPU hours and rendering speed in frames-per-second (FPS) of SA-GS and eleven comparative methods on all eight AOIs in Table 9. Four points could be seen from this table: (i) The conventional NeRF-based methods (Derksen and Izzo, 2021; Marí et al., 2022; Gableman and Kak, 2024) are highly time-consuming in both training and rendering. Even for the fast NeRF-based methods (Müller et al., 2022; Billouard et al., 2024) that employ acceleration strategies, their rendering speed still falls short of most comparative 3DGS-based methods. (ii) The conventional 3DGS (Kerbl et al., 2023), Mip-Splatting (Yu et al., 2024), and EOGS (Aira et al., 2025) exhibit the shortest training time and fastest rendering speed, but perform poorly in satellite image rendering, as demonstrated by the results in Tables 2 and 3. (iii) The proposed SA-GS without the seasonal embedding module requires less training time and achieves faster rendering speed than the full SA-GS. (iv) Our full SA-GS shows moderate performance in both training and rendering efficiency. However, when combining the results in Tables 2 and 3 with this table, it can be observed that SA-GS achieves an effective balance among training time, rendering speed, and accuracy.

5.4. Discussion

According to the reported results in the above subsections, the following points could be revealed:

(1) As seen from Tables 2 and 3, the conventional 3DGS (Kerbl et al., 2023) and Mip-Splatting (Yu et al., 2024) exhibit poor performances mainly because such images are not subject to the pinhole camera model. Although some 3DGS variants (D. Zhang et al., 2024; Kulhanek et al., 2024; Aira et al., 2025; Bai et al., 2025) achieve relatively better performances, the improvements are still limited primarily due to their dependence on the pinhole camera model. In contrast, the proposed SA-GS significantly outperforms all the comparative methods, mainly because it explores the affine version of 3DGS under the affine camera model, which could effectively model the imaging process of line-array satellite sensors, and incorporates the seasonal embedding module and the affine augmentation module to further improve the model performance.

(2) As seen from Table 4, the proposed SA-GS still performs significantly better than other comparative methods under the single-season setting, mainly because the proposed affine view augmentation module helps alleviate the overfitting problem by generating synthetic satellite images as additional training supervision.

(3) As seen from Table 5, the baseline exhibits severely degraded performance, mainly because the baseline is the original 3DGS that is designed for handling plane-array images, and its projection process is subject to the standard pinhole camera model. However, satellite images are generally line-array images that could be modeled by the affine camera model rather than the pinhole camera model. This issue means that the baseline 3DGS could not effectively handle line-array satellite images in theory. By adding AGSM that employs the affine projection, the performance of the proposed method significantly improves. Then, the addition of SEM leads to a significant performance improvement, mainly because SEM helps empower each Gaussian representation to tolerate textural variations among cross-season satellite images by incorporating different types of learnable embeddings. Furthermore, AVAM further improves the model’s performance, mainly because AVAM generates additional multi-view supervisions that help preserve scene details. Since many scene details are correlated with seasons, combining AVAM with SEM allows the model to learn both

scene consistency and seasonal variations, resulting in better performances. Moreover, as seen from the second column of Fig. 5, both the rendered images are black, which could be explained by utilizing the aforementioned discussion: Specifically, since the baseline 3DGS could not effectively handle line-array satellite images in theory, when the 3DGS is trained with such line-array satellite images, the spatial positions of the obtained 3D Gaussians fall outside the visible scene region, leading to the black rendered images whose pixels' colors have to be assigned with the background color (i.e., black in this work).

(4) As noted from Table 7, the experimental results are consistent with the theoretical analysis in the note of Section 3, where higher-degree SH coefficients generally lead to better performance due to their stronger ability in modeling view-dependent appearances. It is also noted that the improvement margin becomes marginal when the degree is no less than 2, mainly because the 2nd-degree SH coefficients already determine a relatively sufficient number of basis functions that could model many common appearances in principle.

(5) As seen from Table 9, the conventional NeRF-based methods (Derksen and Izzo, 2021; Mari et al., 2022; Gableman and Kak, 2024) are highly time-consuming in both training and rendering mainly due to the intrinsic architecture of NeRF. Even for the fast NeRF-based methods (Müller et al., 2022; Billouard et al., 2024), although they employ acceleration strategies to mitigate the inherent inefficiency, their rendering speed still falls short of most comparative 3DGS-based methods, mainly because their rendering pipelines still rely on ray-casting-based volume rendering. The additional computational cost of the full SA-GS compared to its variant without SEM is mainly because the seasonal embedding module introduces an additional dual-embedding MLP to modulate the Gaussian colors, and the modulated Gaussians are then passed into the affine 3D Gaussian Splatting module for an extra rendering process.

6. Conclusion

In this paper, we propose SA-GS, a Season-aware Affine 3D Gaussian Splatting method for satellite image rendering. Under the proposed SA-GS, motivated by the fact that local regions of line-array images could be effectively modeled by the affine camera model, the affine version of 3DGS is explored in the designed affine 3D Gaussian Splatting module for rendering line-array satellite images under the affine camera model. Then, considering that the satellite images captured in different seasons exhibit significant textural variations even for the same region, the seasonal embedding module is designed to empower each 3D Gaussian representation to effectively discriminate textural variations among cross-season satellite images. Additionally, since the number of available line-array satellite images for a given scene is generally limited, which may lead to model overfitting, the affine view augmentation module is designed to synthesize satellite images from different viewpoints as additional training supervision for further boosting the model's performance. Experimental results on eight areas of interest demonstrate that the proposed SA-GS significantly outperforms the comparative methods.

In the future, we will explore more effective strategies for enhancing Gaussian representations to further improve the model's performance in satellite image rendering. Furthermore, we will investigate the potential of the proposed method for 3D geometry reconstruction from satellite images.

CRediT authorship contribution statement

Yihang Xu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Qiulei Dong:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant Nos. 62376269, 61991423, 62472262), Taishan Scholarship Construction Engineering.

Appendix A. Differentiability of affine 3DGS

The key difference between the conventional 3DGS and our affine 3DGS is the projection of the Gaussian's center μ_i and covariance Σ_i to obtain the 2D center μ'_i and the 2D covariance Σ'_i , which is defined in Eq. (6). The gradient of the loss L with respect to μ_i and Σ_i is computed via the chain rule as follows:

$$\frac{\partial L}{\partial \mu_i} = \frac{\partial L}{\partial \mu'_i} \frac{\partial \mu'_i}{\partial \mu_i} = \frac{\partial L}{\partial \mu'_i} M, \quad \frac{\partial L}{\partial \Sigma_i} = M^T \frac{\partial L}{\partial \Sigma'_i} M. \quad (20)$$

Since the gradients $\frac{\partial L}{\partial \mu'_i}$ and $\frac{\partial L}{\partial \Sigma'_i}$ are provided by the differentiable rasterization process as in Kerbl et al. (2023), both μ_i and Σ_i can be optimized accordingly. Moreover, it is noted that the remaining steps in the rendering process of the proposed method are identical to those in the conventional 3DGS, and their differentiability has been verified in Kerbl et al. (2023); hence, the proposed affine 3DGS rendering method is differentiable.

Appendix B. Implementation details of affine 3DGS

It is noted that the proposed affine 3DGS is explored under the affine camera model where the optical center locates at the infinity plane, rather than the pinhole camera model. Accordingly, the viewing range is defined in the proposed method by a parallelepiped (the top and bottom planes sharing identical dimensions, equal to the image size), where the viewing direction is identical for each pixel, instead of a frustum (the basal part of a pyramid formed by cutting off the top by a plane parallel to the base) in the existing 3DGS methods under the pinhole camera model. Then, within the parallelepiped, the 3D Gaussians are splatted onto the image plane under the affine projection defined in Eq. (6) for image rendering.

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